

MPH 2023 Capstone Project

Developing an e-game as a training tool to strengthen public health systems in LMIC settings.

by

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Abstract:

Video games have been used as an interactive education tool in many fields. Most predominantly, the use of games as training, assessment, and evaluation tools is seen in fields of mental health and education in younger age groups. However, there is very little implementation of this kind in the field of public health. This paper explores what a game development process looks like in the field of academia for non-developers with a focus on the challenges, management strategies, and tools for evaluation of outcomes. The game aims to train providers and healthcare workers in systems thinking competencies through an interactive medium. Public health training in many low- and middle-income settings (LMICs) has been inadequate in instilling the significance of systems thinking and collaborative approaches to public health in their healthcare curriculums. The game design is primarily based on an experiential team learning game called “Friday Night at the ER” which is an in-person game focused on systems thinking and management. The gaming industry has seen massive growth in recent times and video games have become more widely accepted as mainstream media for storytelling and entertainment. With low- and middle-income countries having greater access to high-speed internet and declining costs of computing technologies, the popularity of video games has seen rapid growth in countries like India. The methodology explored in this article could help inform future development efforts of games in these growing markets and have an impact on the field of public health.

I. Introduction

a. Background of the project:

Uttar Pradesh (UP) has historically been one of the states in India with significant challenges in the field of healthcare. UP is the most populous state in India with a population of 240 million with over 77% living in rural areas. About 40.5% disease burden is contributed by

Communicable, Maternal, Neonatal, and Nutritional Diseases (CMNND). (*Uttar Pradesh Health Systems Strengthening Project*)

In order to improve the public health outcomes in Uttar Pradesh, the government of UP has recognized the need to improve public health training in institutions and increase core skills of the public health workforce. Under the Uttar Pradesh Health Systems Strengthening Project (UPHSSS) project, the Johns Hopkins Bloomberg School of Public Health (BSPH) has carried out a wide range of initiatives to improve the state's ecology for public health training at the individual, organizational, and systemic levels.

The game development project is part of the Public Health Training and Capacity Building arm of this project. This is in partnership with the Bill & Melinda Gates Foundation, the Government of Uttar Pradesh, All India Institute for Medical Sciences (AIIMS) in Gorakhpur, and the State Institute of Health and Family Welfare (SIHFW) in Lucknow, in India (*Uttar Pradesh Health Systems Strengthening Project*).

b. Systems thinking in healthcare:

The concept of Systems thinking does not have a single origin, but rather it is a method of problem-solving practice that has evolved in the early 20th century through various disciplines including biology, anthropology, physics, psychology, mathematics, management, and computer science (D. H. Peters, 2014). Appropriately, systems thinking is now implemented as a core competency in most MPH and Ph.D. programs (Adam & De Savigny, 2012).

Health System Actors, Functions, and Outcomes

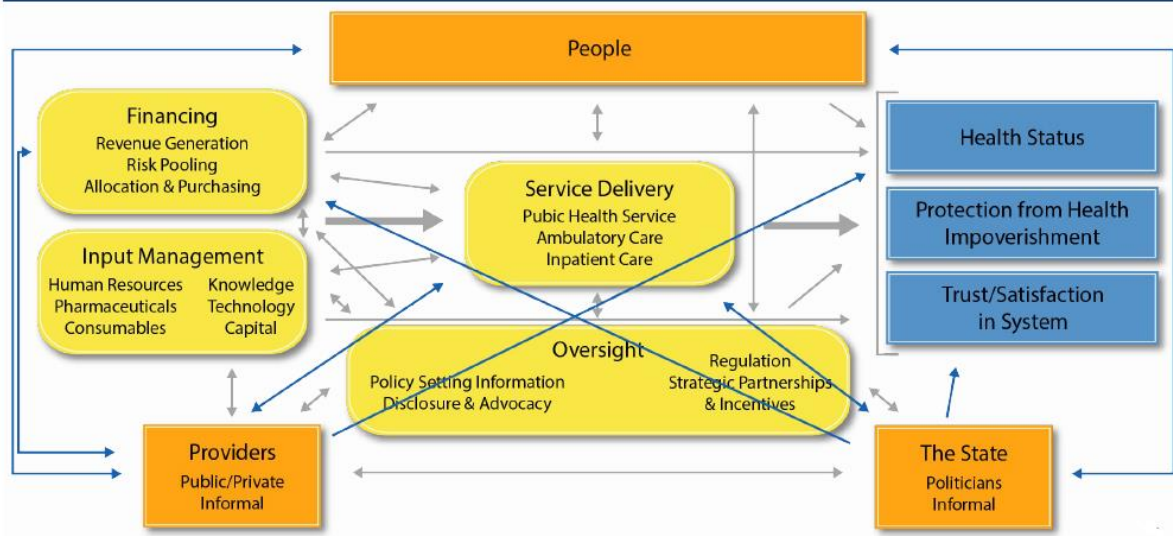


Figure 1. Health systems framework (D. Peters, 2023)

Health systems are inherently Complex adaptive systems that can affect the health of individuals and the community through various components that are often unpredictable and have interdependent relationships with inconsistent outcomes. Traditionally, health systems interventions are developed to focus on individual components of the system (Adam & De Savigny, 2012). It is important to understand these relationships and develop explicit models to guide policies and solutions. Health systems are also ever-changing which makes it essential that models should be scalable to encompass new factors (Leischow et al., 2008).

Adam, et al, in their 2012 paper, called for the need for a paradigm shift in the way health systems problems are approached in LMICs. They especially called for a further need for research in this area and the importance of an enhanced understanding of systems thinking in the public health training in LMICs (Adam & De Savigny, 2012). The focus of this project has been in service of a similar goal to improve the state of public health training using a modern approach in digital education.

c. Video games as training tools:

The term “video game” does not immediately inspire an idea of learning or knowledge. Indeed, historically, video games have been associated with some negative connotations as being unproductive past times and cause for poor grades in children. However, the relationship between science and video games goes back to their very origin (Garris et al., 2002). It was a physicist, William Higinbotham, who is credited for inventing the first video game called *Tennis for Two* in 1950. It was nearly two decades later, in the 1970s, that computer games started being patented and published as a source of entertainment (Tretkoff, 2008).

In their paper, Garris and Ahlers, (2002) described the benefits of harnessing video games as a medium for learning. Their article presented a framework describing the components that make for a game that is engaging and effective as a training tool (Garris et al., 2002). In this paper, we have described the process of designing a serious game aimed at training healthcare providers in systems thinking strategies.

d. Friday Night At the ER:

The initial concept of this game was based on the award-winning systems thinking board game *Friday Night at the ER* (FNER) created by Bette Gardener and published by Breakthrough Learning in 1992. This game has been used as a systems thinking learning tool across various industries and is also part of the curriculum of an international health class at Johns Hopkins Bloomberg School of Public Health.

The game is played a series of rounds, where each round represents one hour in the emergency department. Players take turns moving their pieces around the board to treat patients and allocating staff to provide treatments. They manage their time and resources carefully to ensure that all patients receive the care they need. Throughout the game, event cards are drawn that introduce new challenges or complications. Players must work together

to manage these events while still ensuring the patient load is effectively managed. The game ends after a set number of rounds. The players' goal is to earn the highest team score determined by overall financial and quality measures.

Financial performance is calculated by metrics like the number of ambulance diversions, patients in waiting during each round, the number of patients in the waiting area between departments, and extra staff utilized. These metrics have a fixed multiplier assigned to them which gives a total cost to the hospital. Quality measures have similar metrics but have different multiplier values which give a total score of quality errors. The multipliers are not presented to the players outright, but players have the ability to ask for them. This is the hidden mechanic aspect of the game which effectively promotes out-of-box problem solving and data-driven thinking.

FNER is designed to emphasize the importance of collaboration, communication, and careful resource management to provide the best possible care to patients. By mimicking the high-stress environment of an emergency department, the game also encourages players to develop skills in systems thinking by stimulating mental models and then challenging them, such as balancing feedback loops and managing common resources.

The true value of *FNER*'s teachings lies in how the structure of the board and presentation of the rules imply certain constraints (utilization of ludo narrative dissonance). It is up to the player to think outside the box, try to find novel solutions outside of what the game mechanics present, and proactively ask for information.

II. Literature review

In recent years, video games have grown into a substantial form of entertainment and education in mainstream media. Increased computing power, decreasing cost of hardware, and improved internet access have allowed developers to put video games within reach of a

larger audience. Growth has been significant, especially within the realm of mobile and browser-based games which are relatively less demanding than console-based games (Abdulla, 2021).

A systematic review published in 2018, looked at the utilization of video games in medical education and the findings of their effectiveness as learning tools (Gorbanev et al., 2018). From the results of this study, it was noted that games are more effective as complementary training tools and adjuncts to classroom teaching rather than teaching tools in and of themselves. There was also the consideration that access to medical games is limited.

Several serious games have been developed in the healthcare space, however, there is limited implementation in the public health field. The *Logo educational video game*, designed by Feurzeig, Papert, and Solomon in 1967, can be considered the first educational video game. *The Oregon Trail* published in 1971 was in use as an educational game in schools for nearly 30 years (Breien & Wasson, 2022).

A notable example of a public health serious game is *Point of Contact* which was designed during the COVID-19 pandemic to improve understanding of preventive measures among children (Hill et al., 2021). Other games in the public health space are *Crystal Island*, which has proven to be an effective learning tool in scientific study (Breien & Wasson, 2022), as well as *Outbreak* which allows participants to role-play and put their knowledge of controlling an infectious disease epidemic in a major urban area to the test (Connor & Phillips, 2007).

A notable example of a video game used as a treatment intervention is *EndeavorRx* which became the first video game to receive FDA approval as an adjunctive treatment for ADHD (FDA, 2020).

While there are a few games related to public health competencies, this remains a highly untapped market in the field of education. Additionally, as of the writing of this paper, there seem to be no games specifically focusing on systems thinking in the public health setting. The goal of this project is to fill that gap and build an effective tool for enhancing public health education in LMICs.

This paper highlights the management and leadership challenges and solutions of co-design a game being led by public health researchers who do not have a technical background in game development. This paper will also discuss the various stages in the design process and the path of the project moving forward.

III. Methods

a. Why design a video game?

The project team has had first-hand experience with FNER's model and its benefits in the reinforcement of systems thinking education. However, we identified certain limitations with the game. Primarily, this being a board game, it required players to have and manage several game kits with many pieces. Apart from licensing being expensive, it is hard to procure copies of the game in rural LMIC settings. This poses a significant barrier to access for such learning tools. Additionally, a board game is limited in the scope of complexity and mechanics since it relies on manual calculations.

A video game allows for deeper narrative and computational complexity which would be a more rewarding experience. It would also be easier to distribute copies of the game over the internet to rural areas which have limited access. Once the basic mechanics have been developed, there is the possibility to scale the game design to accommodate a larger scope. It is possible to develop different modules that focus on different areas of public health and even to model the game to different cultural and socio-political contexts.

b. The Team:

The internal game development team (the Team) was led by a faculty member with experience in a serious games lab and consisted of a scientific staff member and two graduate students. This team was responsible for developing the game principles, refining the original design, and providing subject matter expertise around public health principles and systems thinking. The team later engaged a professional game designer in the capacity of an external consultant to advise on the technical aspects of game design and development and help establish game design principles to reflect the learning objectives and scenarios. The product owners were two senior faculty members who lead the primary UPHSS project. They provided oversight and approval of interim design and were also responsible for the preliminary concept development. The coding and implementation of the concept will be done in with Juego Studios, India.

c. Agile Methodology:

The internal development team chose to implement an agile management methodology for this project instead of the traditional waterfall method for several reasons. The agile method has proven to be more effective in co-development and rapid iterative designing of innovative technologies in healthcare(Kannan et al., 2017). The agile method breaks the development process down into small, iterative cycles, or sprints. This allows the team to quickly adapt to changes in the requirements and to deliver working software early and often. A primary benefit is that it allows the team to get feedback from users more frequently. This feedback can be used to improve the game and to ensure that it meets the needs of the target audience. The agile method also allows the team to be flexible and to adapt to changes in the requirements on the go. This is important in the scientific and public health field, where standards can change rapidly due to new research or policy changes. Given the nature of this

project particularly, the agile method helped the team adapt to changing goals and design constraints rapidly. Thus, making the turnaround time for each game design principle as short as possible. The key roles in this methodology were the product owners, in this case, the senior faculty members; the development team responsible for iteratively designing the game principles and framework for the game model; and the stakeholders consisting of end users and experts from the target community who provided valuable perspectives on aspects of the design and supplied context on culturally specific questions.

d. Original game concept and refinement:

The game was originally conceived in July 2020 by the product owners, who brainstormed together with the UPHSS team to design a game concept that reflected some of the principles of FNER but focused on public health competencies rather than a healthcare delivery setting and an engagement contract was signed with Juego Studios in August 2021. This original approach was to develop a fully thought-out framework before beginning the game development process with the studio to ensure the game that was developed reflected the scientific team’s conception exactly. This presented a challenge to the original timeline. A faculty public health scientist with serious game development experience was brought onto the project in July 2022, and the graduate research assistants joined in October 2022 (the Team). The Team began to refine the original concept, using weekly meetings in line with the agile framework to keep frequent checks on the progress and iteratively problem-solve.

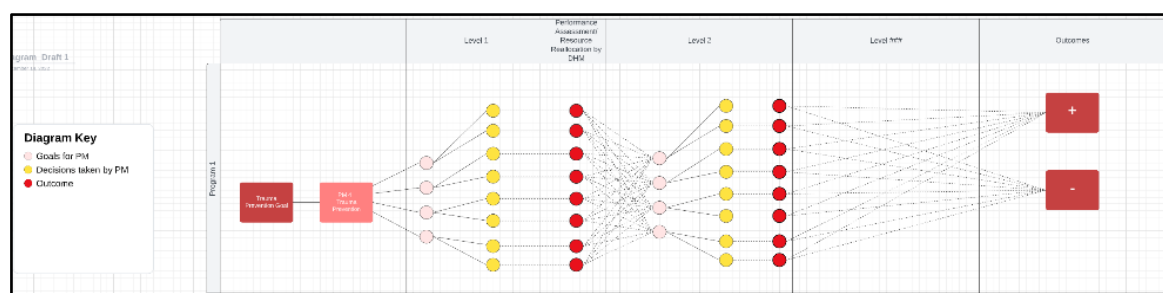


Figure 2. Game model 2 – Game design choices mapping

Game model 1

This first iteration of the game model (Fig 2) was refined by the Team based on the primary design from the product owners without engagement from subject matter experts with video game design expertise (ie, videogame designers). This model was designed with a five-player team with four players being in the role of a Program Manager and one in the role of district health manager (DHM). In this model, all the players (except the DHM) had the same responsibilities and actions. Each player would focus on their own program area (e.g., maternal and child health or noncommunicable disease) rather than a distinct function of public health. The gameplay in this model had players decide on health program targets based on an initial community health assessment. Then the game presented them with a set of actions that they could take each round. The choice of action made would guide the game's progression.

This model was presented to the game development studio in December 2022, and they requested a deeper cache of information in designing each set of metrics and choices. As the Team started mapping out the choices to be presented and delving deeper into the literature to address these questions, the complexity of all the possible pathways became more apparent. One approach to tackle this was using a qualitative research methodology to index information from relevant literature into a database. We used *Atlas.ti* for this purpose and constructed a detailed database from public health literature around LMICs, focusing on Uttar Pradesh and the work of the UPHSS project. It was recognized that the project did not have adequate resources to execute this level of complexity in designing the game. It became necessary to reassess the design goals and try to simplify the scope of the game.

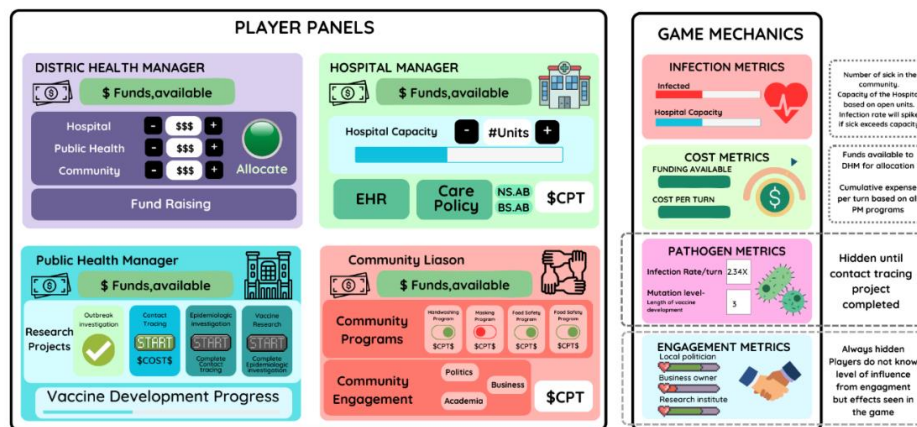


Figure 3. Game model 2 – Game mechanics framework

Game model 2

A second iteration of the game was designed by this author (Figure 3). This author identifies as a gamer and has extensive experience playing video games and has clinical background of working in India. It was this joint experience in video games and healthcare that was useful in designing this game model. The scope of the game was limited to an epidemic outbreak in the community and a new design was created which was based on using a point-based system rather than a choice-based system to determine health outcomes. In this model, each of the players was assigned a distinct role pertinent to an aspect of the health system. Each player had a limited number of distinct actions during each round and the outcome of those actions would be judged by a point-based system. A formal stakeholder engagement phase was also implemented at this time for more feedback on the game.

In January 2023, the design consultant was brought onto the team to provide a better perspective on codesigning with stakeholders and to guide the process leading up to re-engagement with Juego. Certain principles from the eLuna codesign framework were borrowed while redefining the design process to be more streamlined and driven by stakeholder inputs. The eLuna framework is a tool for educators and game developers to

work together and co-design a video game in a structured way. The framework involves an initial phase where educators define the curricular content, learning objectives, learner demographics, and learning situation. This is followed by a co-design phase where the co-designers produce a game description in table format. The co-specification phase then results in a narrative blueprint, which is used in the final development phase to create the game. The eLuna framework uses a visual language based on icons, colors, and written descriptions to facilitate the co-design and co-specification phases, allowing for flexibility during the design phase and strict adherence to the blueprint during the specification phase (Breien & Wasson, 2022). Unlike waterfall management, the agile methodology does not require planning out detailed steps of the entire product pipeline. The iterative nature of frequent, and quick feedback has made this style most effective in implementing this game model.

The product owners pointed out that this model did not provide much flexibility to design any additional scenarios or modules if it was so desired. It was also decided that the game would provide more value to players if the context was focused on health issues more prevalent in the state of UP, so it would be good to move away from the epidemic context.

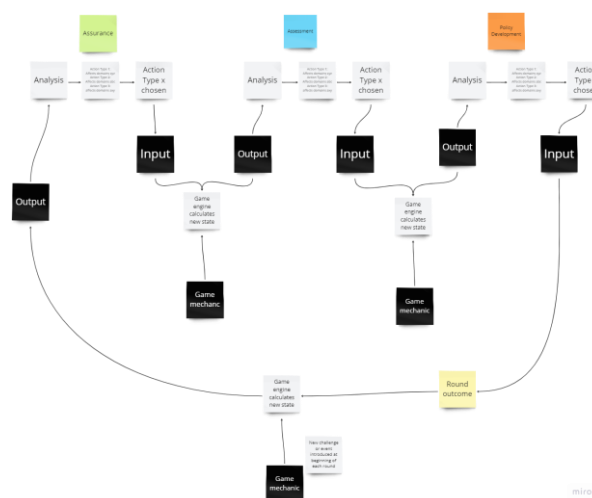


Figure 4. Model 3- Design based on the CDC public health framework

Game model 3

This was the most challenging phase of the project as there seemed to be a disconnect between the Team's expertise in public health and education and the technical aspects of meeting stakeholder preferences and product owner requirements to implement in the game design. The agile methodology was especially useful in this situation. It allowed the Team to quickly address emergent problems and iteratively problem-solve.

A shared idea board (Miro) was introduced by the design consultant which was used for brainstorming and initial concept mapping through regular sessions of discussions and shared problem-solving with the BSPH team. Generally, these tools are utilized more in the knowledge development space but the team was able to quickly adapt to it and demonstrate how this tool was effective in developing the game concept and functioned as a shared workspace.

Keeping in line with the objective of the game reflecting real-life scenarios, the first step was to map systems relationships in public health. Key domains of public health were identified and their interconnected relationships were mapped using Miro.

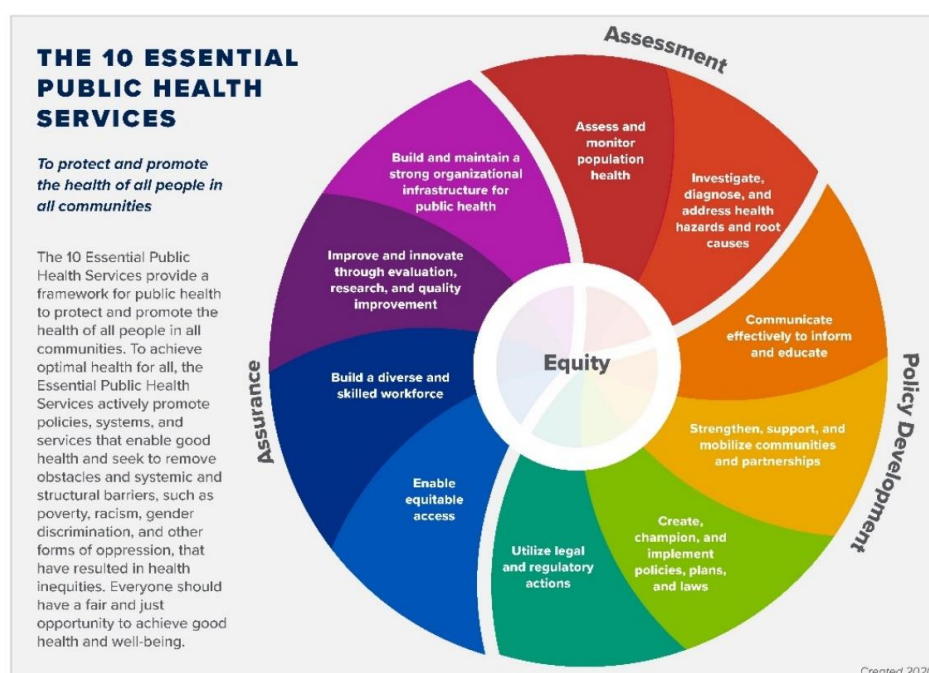


Figure 5. CDC Public Health Services Framework (10 Essential Public Health Services - Public Health Infrastructure Center, 2020)

The CDC framework describes a cyclical arrangement of public health services (Figure 5). This framework was revised in 2020 and is being used in its current form. The framework describes ten public health services, broadly classified under three overarching areas of Assessment, Assurance, and Policy Development (*10 Essential Public Health Services - Public Health Infrastructure Center, 2020*). This inspired the team to consider a circular structure for each round.

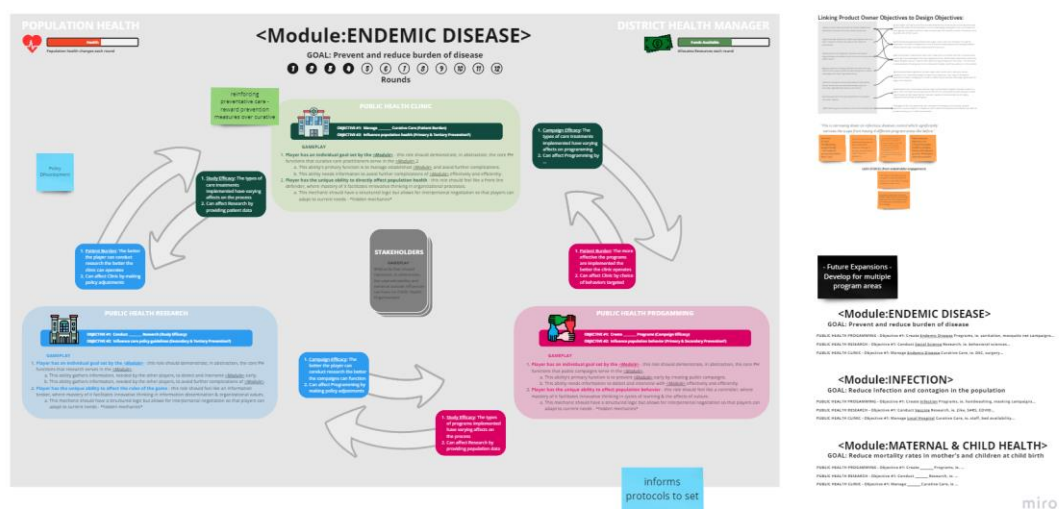


Figure 6. Game design framework showing the cyclical nature of gameplay mechanics

This very closely describes the interconnected nature of all elements of the public health system within the three overarching areas. Each round updates the game state and player choices have a collective impact on the health outcomes for that round (Figure 4). For designing the framework, it was necessary to go deeper into the specific functions of each of these overarching areas. The team, in collaboration with the design consultant, recreated this framework with an in-depth mapping of the functions of the public health system for specific public health challenges. The Team started developing scenario models based on this framework. We selected specific public health issues and mapped how each area of public health would contribute to the management of the issues, the actions taken under each domain, and how those actions would be interconnected. Keeping in line with the agile methodology, this was an iterative process led by the designer with frequent brainstorming

and requests for scientific content that helped inform the design principles. It led to the creation of storyboards and action flowcharts, documenting the issue from diagnosis and assessment to policy creation and assurance.

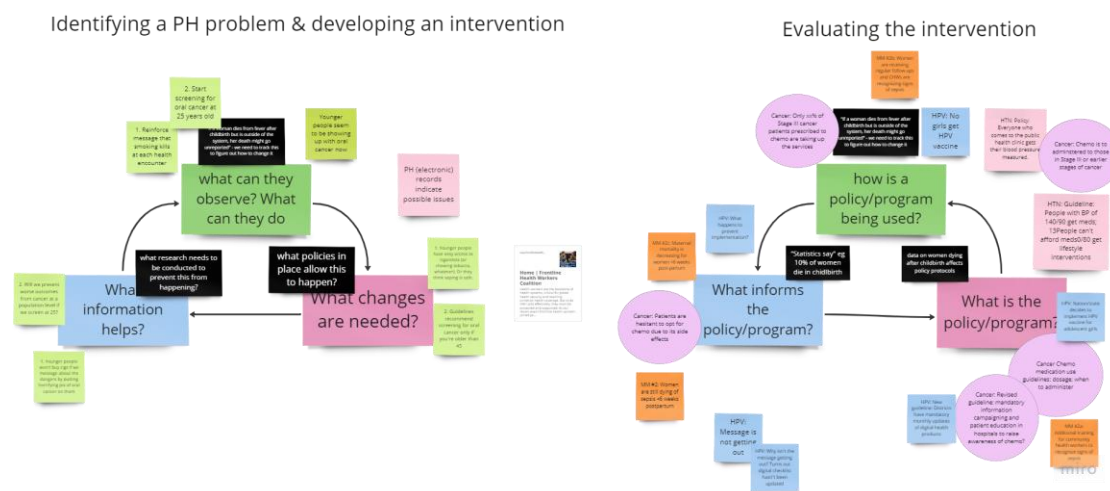


Figure 7. Public health systems mapping

Once these relationships were mapped, a game model was developed to describe how the game mechanics would reflect the complexities of the real world, defining the interactions between players and the game environment.

IV. Final Design:

The project is now close to re-engaging with Juego Studios and beginning the formal development phase. As of the time of this writing, we are awaiting final approval from the product owners on the game principles (presentation scheduled for May 2), and engagement with Juego Studios is expected to begin in 1-2 weeks. It is also worth noting that, while this will mark the end of the design phase, the development phase will continue to be iterative and the team will closely collaborate with the development studio.

While the design borrowed parts from various frameworks, the primary methodology to generate content for the plan has developed into a method that involves regular and frequent

stakeholder engagement. This generative co-design framework was implemented to engage stakeholders as if they are members of the team.

a. Specifications of the game:

i. Target users: The target users are healthcare/public health workers in UP, India. Primarily the focus would be on students and professionals in the early career stage. However, the involvement of more experienced practitioners would provide a deeper context and understanding of the game.

ii. Learning objectives: The primary objective is to encourage systems thinking among public health professionals through a narrative-based digital learning environment. It should be noted that the goal is not to teach public health competencies but rather to reinforce systems thinking principles from classroom teaching using an interactive platform. This provides the opportunity for participants to apply their concepts, and to experience how the outcomes in a health system vary based on shared decision-making.

Specifically, the intent is to stimulate the players to think about mental models, prompt deeper consideration of how seemingly good choices can have far-reaching impacts on other aspects of the system, and ultimately promote a discussion of “wicked problems” that are prevalent in the system and the importance of tackling these issues in the real world. Wicked Problems are defined as those problems in a policy or planning systems that are complex, persistent, and require innovative solutions to solve (Rittel & Webber, 1973).

iii. Design constraints:

- The most important constraints in the project have been time and resources available. It was expected that the project would take 2 years to develop a finished product. Significant person-hours and resources were utilized to create the original concept of the game, however, there was a gap in the resources available for the scope of the

original design, and the concept had to be refined and the scope limited to fit constraints better.

- The game must be simple enough for utilization by individuals not familiar with computer games. The design and controls will have to be intuitive enough to learn and master within a short time.
- The entire playtime must be limited to 1-2 hours which would be the typical length of a training session.
- The context of the game would have to be familiar to the players for learnings to be relevant.
- Game elements should reflect real-life public health scenarios.
- Diversity, Equity, and Inclusivity consideration to be upheld.

iv. Platform: The game platform was an important consideration for success. Modern consoles would be unsuitable due to high costs and lack of availability in rural India. Based on user input, it was identified that the most accessible platforms would be a PC or mobile device game. The design would have to be light, and capable of swift running without demanding processor hardware. Ideally, this would be a browser-based game that would be logged into from any device although a stable internet connection would be required for efficient gameplay.

V. Setup: The game would be multiplayer, similar to FNER, and intended to be played in teams of 3-4 players. Each player would oversee a discrete aspect of the public health system: Assessment, Assurance, and Policy Development.

vi. Goal: The team must work together towards a common goal of solving a public health problem. The team with the best outcomes in the session would win.

b. Mechanics:

The eLuna framework describes a scale of freedom of player behavior and choice which is allowed within the game. The narrative pole allows the player more freedom to explore the environment and be immersed in the story, this comes at the expense of flexibility in the kinds of actions players can take and having limited choices. On the Ludic pole the game allows for more functionality, prioritizing mechanics and actions of gameplay but not providing much in the way of narrative depth (Breien & Wasson, 2022).

For this game, it is most appropriate for the game model to be leaning toward the Ludic pole. This is because narrative is not the key element of the game, rather providing players with high flexibility to make varying choices has more value. It is worth noting that the game is agnostic to any specific genre. While the closest classification of this game could be made as a strategy game, it is not necessarily a primary consideration. The specific genre of the game would be decided by the development team based on the key principles to be included.

The primary element of gameplay mechanics is the interdependent impact of systems (as represented by the three players) on the health outcome of populations. The setting of the game was designed to be a fictional community in rural India, like the predominant populace of UP. All characters and objects in the game will be modeled for this setting.

Like *FNER*, the learning value of the game is driven by ludo narrative dissonance. The setup and framework of the game suggest that each system of the game is a separate entity and players must focus on advancing individual agendas. This is like real-world scenarios where healthcare agencies are not cooperating. Through the outcome of the gameplay, and by employing the principles of ludo narrative dissonance, players will be led to the understanding that the best outcomes are achieved when all components of the system function in harmony with one another and cooperate to solve problems. They will gain an understanding of resolving competing agendas to achieve optimal outcomes, and strategies in

the management of limited shared resources. Users are expected to be able to translate these learnings to reinforce systems thinking principles.

c. Scoring:

As discussed earlier, the scoring criteria in *FNER* use measures of quality and financial performance as metrics for calculating the outcome of each team. There is also the hidden mechanic of players being able to ask for information about the performance metrics. This creates a moment of surprise and makes for a rewarding learning experience through the principle of ludo narrative dissonance.

Following a similar strategy, the scoring criteria for this game are modeled after standard systems quality measurement frameworks.

Outcomes of the health systems would be evaluated using a point-based system applied to domains under each dimension in the traditional quality assessment framework (Deroose et al., 2002). The planned metrics of measurement in each quality dimension would be as follows:

Structure: Equity, Access

Process: Efficiency, Population Centered

Outcome: Effectiveness

This would help players relate the outcomes to real-world scenarios better and also provide a viable scoring scheme for each team. Player actions will have a cumulative effect on the score in each dimension of quality indicators. These scores will be aggregated for the final score of the system. Teams will be ranked based on the final score in each session and the team with the highest score will be the winner.

Similar to *FNER*, there would be a deeper hidden layer of metrics for this assessment and players will have the ability to seek out this information if they employ innovative approaches.

d. User Experience:

During the development of this project, stakeholder feedback has been given a high priority in informing design decisions. The stakeholder engagement process is beyond the scope of this article, however, the inputs from potential users have been integrated into the design of the game. As the project begins to move into the development phase with *Juego*, we will continue stakeholder engagement during this phase. The development will also be iteratively informed by stakeholder feedback.

e: Testing: Testing of the game will be in line with standard industry practices with two primary stages of testing. Other serious game development pipelines have demonstrated effective use of this form of user testing (Johnsen et al., 2018).

i. Alpha testing: Alpha testing primarily tests for usability and functionality. At this stage, the game will be tested internally by the developers to ensure all elements of the game are working properly. This will be vital in identifying bugs in the functionality. This stage is expected to last for two weeks of iterative debugging and patching to ensure the game build can perform as intended.

ii. Beta testing: After the alpha build is completed, the game will be open for beta testing. Ideally, at this stage, volunteers will be chosen for beta testing. The primary goal of this testing stage is to collect user feedback on aspects of gameplay, understanding of the mechanics, and user experience. Bugs identified during beta testing will not be iteratively patched. All debugging and patching will be conducted at the end of the beta testing phase after feedback from users has been obtained and analyzed in detail. This stage is expected to

last for a shorter period, typically allowing for a few game sessions and immediate feedback following the sessions.

V. Evaluation Plan

An important managerial consideration of this project is the monitoring and evaluation of the effectiveness of the learning objectives in the context of the broader systems-strengthening project. The ultimate indicators of success for the UPHSS project will be based on the WHO framework for assessing health systems' strengths. This is not within the purview of this paper. The specific outcomes of the game development and utilization will be measured as described below.

a. Evaluation of learning objectives:

The measurement of a thinking process is very challenging. Specifically, the challenges are around the potential for bias in reporting and how the instrument would favor certain knowledge over others (Grohs et al., 2018).

We will use the 20-item systems thinking scale as a measure of understanding among players. This is a validated scale that measures the *systems thinking construct of system interdependencies in the context of quality improvement* (Dolansky et al., 2020).

Key learning objectives to be measured:

1. Systems thinking construct: How well did players understand the interconnectedness of systems.
2. Challenging mental models: How do players stimulate the building and challenge traditional mental models and how this drives player behavior.
3. Data-driven modeling: How well do players implement data-driven methodology.
4. Innovation: How efficiently were players able to solve complex problems in game.

5. Collaboration and communication: How well did players collaborate on the team.
6. Application: How well do players understand the transference of these skills to practice.

This will be done through a comparison of pre-and post-intervention questionnaires.

Participants will take a quiz with questions based on systems thinking principles and implementation in public health before the game session. Immediately after the game session, the participants will retake the quiz. Scores from both questionnaires will be compared to assess the improvement in understanding of systems thinking concepts.

b. Disengagement evaluation:

Disengagement is defined as a loss of interest in players to interact with the game. This would be categorized as one or more of the following activities. These criteria were adapted from a similar evaluation in a different study (Rowe et al., 2010):

1. Spending too much time in areas of the game where there is no value.
2. Interacting with random objects of the game which provide no additional value.
3. Randomly choosing options without thoughtful consideration.
4. Deliberately choosing incorrect responses to end the interaction early.
5. Attempts to game the systems by deliberately selecting incorrect responses to prompt the game to give correct responses.

Disengagement is detrimental to learning and would help inform decisions in improving design and mechanics.

c. Monitoring and Evaluation plan: The plan for monitoring and evaluation of the game will be iterative given the nature of the project, with additional feedback from stakeholders and users after game sessions have been implemented.

The following measures will be used to evaluate the outcome of completing the training session. Ideally, these evaluations will be carried out monthly.

Indicator	Metric	Measurement
Systems Thinking		
Systems thinking construct	Systems Thinking Scale (STS)	Pre-Post Difference
Data-driven methods	Learning objectives quiz	Pre-Post Difference
Problem-solving	Problem-solving score	Game-Analytics
Collaboration	Internal communication frequency	Game-Analytics
Application	Learning objectives quiz	Pre-Post Difference
Engagement		
Focus	Time spent in unproductive interactions	Game-Analytics
Disinterest	Choosing non-productive actions on purpose	Game-Analytics
Cheating	Attempts to game the system	Game-Analytics
Initiative	Asking the facilitator for additional information	Facilitator report

The objective is an improvement of systems thinking understanding among healthcare workers in India through a virtual serious game learning environment as the intervention.

VI. Discussion

In this paper, we have considered a novel approach to public health education with the use of a serious game. We also discuss what the development process of such a game in an academic setting looks like and the technical and managerial challenges that go along with it.

Video games can be tailored to specific audiences and learning objectives, which can make them more effective as training methods. They can also be used to mimic real-world scenarios, which can help learners to develop the skills they need to make informed decisions in complex situations. Games can also promote collaboration and teamwork, which are essential skills for addressing public health challenges in LMIC settings.

In this project, the agile methodology of management has been most effective in driving the development of our game. The team was able to address challenges with iterative solutions and frequent feedback, ensuring that all issues were managed in the shortest time possible. In the case of this project, designing the framework presented significant challenges due to the scope of the original game concept and implementation strategies. These were addressed by reducing the scope of the game for the first iteration and building the mechanics by adopting some principles from the eLuna framework.

The success of this game will depend on the evaluation of outcome measures, which in the current design will indicate if the game was able to improve the retention and implementation of systems thinking processes.

Due to shifting timelines and the need to change the scope of the game, the project has not yet been able to move into the game development phase. While this is a setback in the project pipeline, it ensures that the team remains responsive to stakeholder input and helps to create a product that would provide value to the end users.

Following the current framework, we expect this game to be able to provide a virtual environment of sufficient complexity that reflect conditions similar to those in UP. The players would be able to test their theoretical understanding of public health and systems thinking and see how their choices affect the outcome of the game.

Ideally, this project should also lead to increased utilization of other such innovative tools in learning and training. With the recent advances in artificial intelligence (AI), the development of games is likely to become much simpler. There are already AI tools capable of writing complex code based on lingual prompts. AI image generators can create sophisticated images and assets. Soon it might be possible for educators with no game development background to be able to build entire games such as this without the need for any coding. For instance, the

below illustrations of a “rural Indian setting” were generated entirely by an AI generator, *Leonardo.AI* using a simple prompt. These could serve as potential high-quality game assets with very little effort. (Figures 8 and 9)



Figure 8. AI-generated images as game assets



Figure 9. AI-generated images as game assets

Regardless of the future of AI, video games are an untapped resource for education and engagement especially among the younger populations, and should be used for more effective communications and training.

VII. Conclusion

The technical and managerial challenges of developing such a game in an academic setting among scientific teams with strong subject matter expertise but limited experience in making video games as demonstrated in this project can serve as an example for similar projects in the future.

Development of such tools can be expensive which can limit their accessibility, especially in LMIC settings without the presence of funders. They also may not be effective for all learners, and some learners may prefer traditional training methods based on the topic of study.

Despite these limitations, video games have the potential to improve public health quality and outcomes in LMICs. The development of a serious game to train public health workers in systems thinking is a complex and challenging task that requires careful planning. Several factors need to be considered, including the target audience, the learning objectives, the game mechanics, and the budget.

Through the development process, we learned that serious games have great potential as training tools, particularly when it comes to addressing complex scenarios and promoting collaboration and teamwork. We also learned that using an agile approach to management was effective in driving the development of the game, allowing for iterative solutions and frequent feedback that ensured issues were managed in the shortest time possible.

The project manager must have a strong understanding of the healthcare field and the needs of public health workers and must be able to set clear goals and objectives, communicate effectively with the team, and resolve conflicts.

The success of this game will depend on the evaluation of outcome measures, which will indicate if the game was able to improve the retention and implementation of systems thinking processes. While the project has not yet moved into the game development phase, updated timelines and improved scope of the game ensures that the team remains responsive to stakeholder input and the product will provide value to end-users.

Looking to the future, we believe that video games are an untapped resource for education and engagement, particularly among younger populations. With the recent advances in

artificial intelligence, the development of games is likely to become much easier and faster, allowing educators with no game development background to build games on their own. We hope that this project will lead to an increased utilization of such innovative tools in learning and training, particularly in LMIC settings where resources may be limited.

Overall, we hope that this paper has contributed to the ongoing conversation about the potential of serious games for public health education and has highlighted the importance of agile management in the development of such games. We look forward to seeing the impact of this project and the future direction of serious game development in public health education.

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